

Codeforces and Polygon may be unavailable from [August 20, 16:00 \(UTC\)](#) to [August 21, 1:00 \(UTC\)](#) due to technical maintenance. ✕

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E. Nearest Tour Destination

time limit per test: 1 second

memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

You are given an **undirected unweighted** graph with N nodes and M edges. The nodes are numbered from 1 to N . The graph contains no self-loops and no multiple edges.

There are S sources and Q destinations. For each destination node, find the length of the shortest path from any source node to that destination. If a destination is unreachable from all sources, output -1 .

Input

The first line contains four integers N, M, S, Q

$1 \leq N \leq 2 \times 10^5$, $0 \leq M \leq 3 \times 10^5$, $1 \leq S \leq N$, $1 \leq Q \leq N$ — the number of nodes, the number of edges, the number of source nodes, and the number of destination nodes.

The next M lines will contain two integers u_i, v_i ($1 \leq u_i, v_i \leq N$) — denoting there is an edge from node u_i to node v_i .

The next line contains S ($1 \leq S_i \leq N$) integers representing the source nodes, and the final line contains Q ($1 \leq Q_i \leq N$) integers representing the destination nodes. A node may appear both as a source and as a destination.

Output

The output should consist of Q integers separated by spaces. The j -th integer denotes the length of the shortest path from any source node to the j -th destination node. If no such path exists for a destination node, print -1 for that destination. A node may be both a source and a destination, in which case the answer for that destination is 0.

Examples

input

Copy

```
8 6 2 4
1 2
2 3
4 5
6 7
7 8
2 6
1 6
3 5 6 8
```

output

Copy

```
2 -1 0 2
```

input

Copy

```
18 17 4 10
1 2
```

CSE221 Assignment 06 Summer 2026

Contest is running

3 days

Contestant


 → **Submit?**

 Language: Java 21 64bit

 Choose file: Choose File No file chosen


Submit

```
2 3
3 4
4 1
3 5
5 6
6 7
8 9
9 10
10 8
10 11
11 12
9 13
13 14
15 16
16 17
17 15
15 1 6 8
14 3 10 7 1 12 11 5 18 16
```

output

Copy

```
3 2 1 1 0 3 2 1 -1 1
```

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The only programming contests Web 2.0 platform
Server time: Aug/20/2026 17:15:46^{UTC+6} (i1).
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